

studentGrade.py

```
1  # function for finding letter grade
2  def letter_grade(score):
3
4      # score must be between 0 and 100
5      if score < 0 or score > 100:
6          raise ValueError("Score should be between 0 and 100")
7
8      # checking grade
9      if score >= 90:
10         return "A"
11     elif score >= 80:
12         return "B"
13     elif score >= 70:
14         return "C"
15     elif score >= 60:
16         return "D"
17     else:
18         return "F"
19
20
21 # function to analyse student data
22 def analyse(students):
23
24     # empty dictionary for storing result
25     results = {}
26
27     # looping through students
28     for name, score in students.items():
29
30         # storing score and grade
31         results[name] = {
32             "score": score,
33             "grade": letter_grade(score)
34         }
35
36     return results
37
38
39 # function for showing summary
40 def summary(results):
41
42     # getting all scores
43     scores = [data["score"] for data in results.values()]
44
45     # calculating average
46     avg = sum(scores) / len(scores)
47
48     # finding highest scorer
49     highest = max(results, key=lambda k: results[k]["score"])
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50
51     # finding lowest scorer
52     lowest = min(results, key=lambda k: results[k]["score"])
53
54     # dictionary for grade count
55     counts = {}
56
57     # counting grades
58     for data in results.values():
59
60         grade = data["grade"]
61
62         counts[grade] = counts.get(grade, 0) + 1
63
64     print("\nClass Average :", f"{avg:.2f}")
65     print(f"Highest Score : {highest} ({results[highest]['score']}")
66     print(f"Lowest Score : {lowest} ({results[lowest]['score']}")
67
68     print(
69         f"Grade Counts : "
70         f"A={counts.get('A', 0)}  "
71         f"B={counts.get('B', 0)}  "
72         f"C={counts.get('C', 0)}  "
73         f"D={counts.get('D', 0)}  "
74         f"F={counts.get('F', 0)} "
75     )
76
77
78     # student data
79     students = {
80         "Alice": 92,
81         "Bob": 78,
82         "Carol": 85,
83         "Dave": 61,
84         "Eve": 55,
85         "Frank": 99
86     }
87
88     # handling invalid score error
89     try:
90
91         # analysing data
92         results = analyse(students)
93
94         # printing each student result
95         for name, data in results.items():
96             print(f"{name:<8}: {data['score']} → {data['grade']}")
97
98         # showing summary
99         summary(results)
100
101     except ValueError as e:
102

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```
103     # if invalid score comes
104     print("Error:", e)
```